

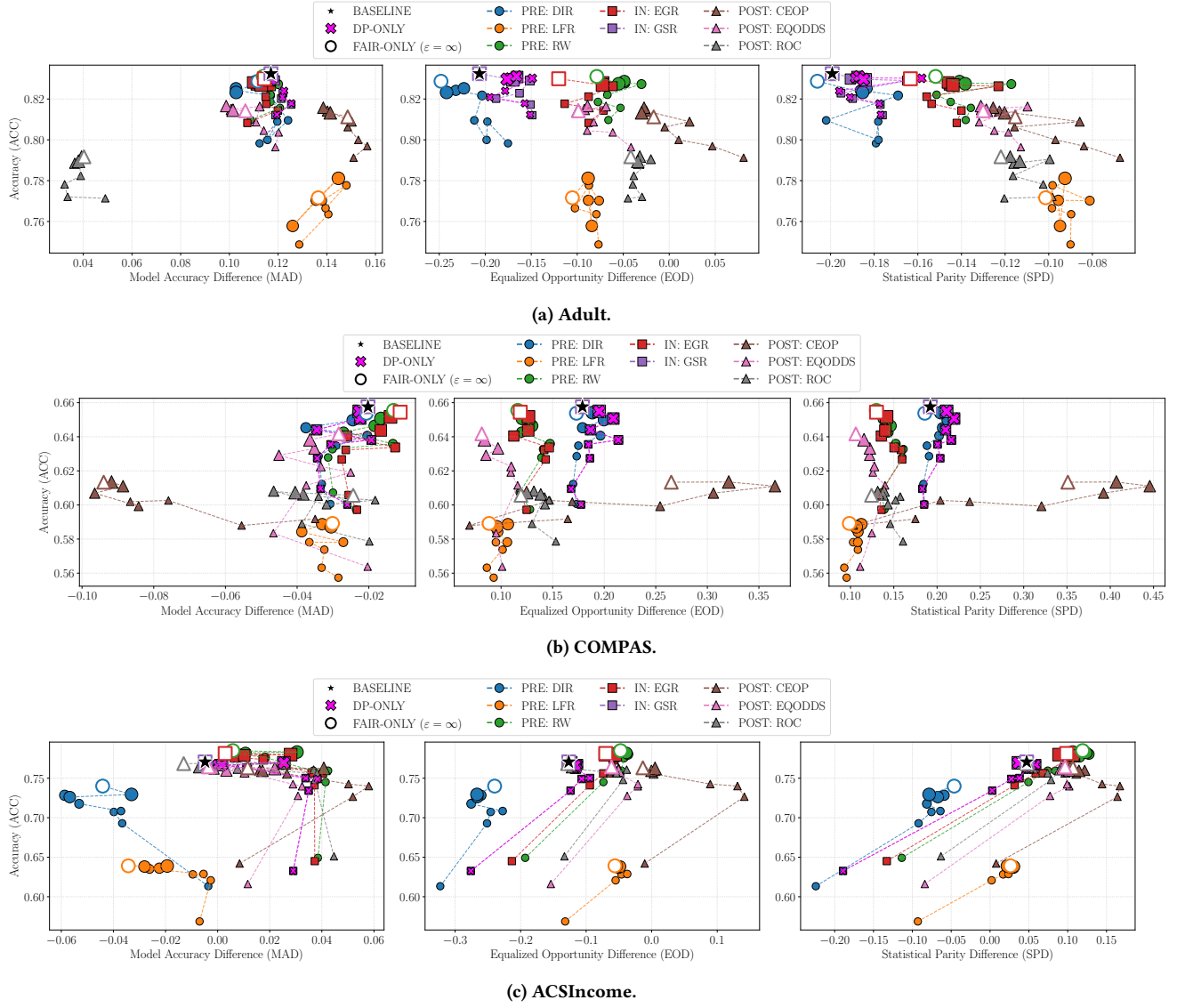


Figure 16: Accuracy–fairness trade-offs under the AIM synthesizer across three real-world datasets. Each subfigure reports accuracy (ACC) versus three fairness metrics (MAD, EOD, SPD) for each fairness intervention applied at the pre-, in-, and post-processing (with DP calibration set) stages. Marker size encodes the privacy budget ϵ for both DP-only and DP+Fair settings, while hollow markers indicate the Fair-only setting ($\epsilon = \infty$). The Baseline setting (no DP and no fairness intervention) is indicated by the $\star$ symbol.

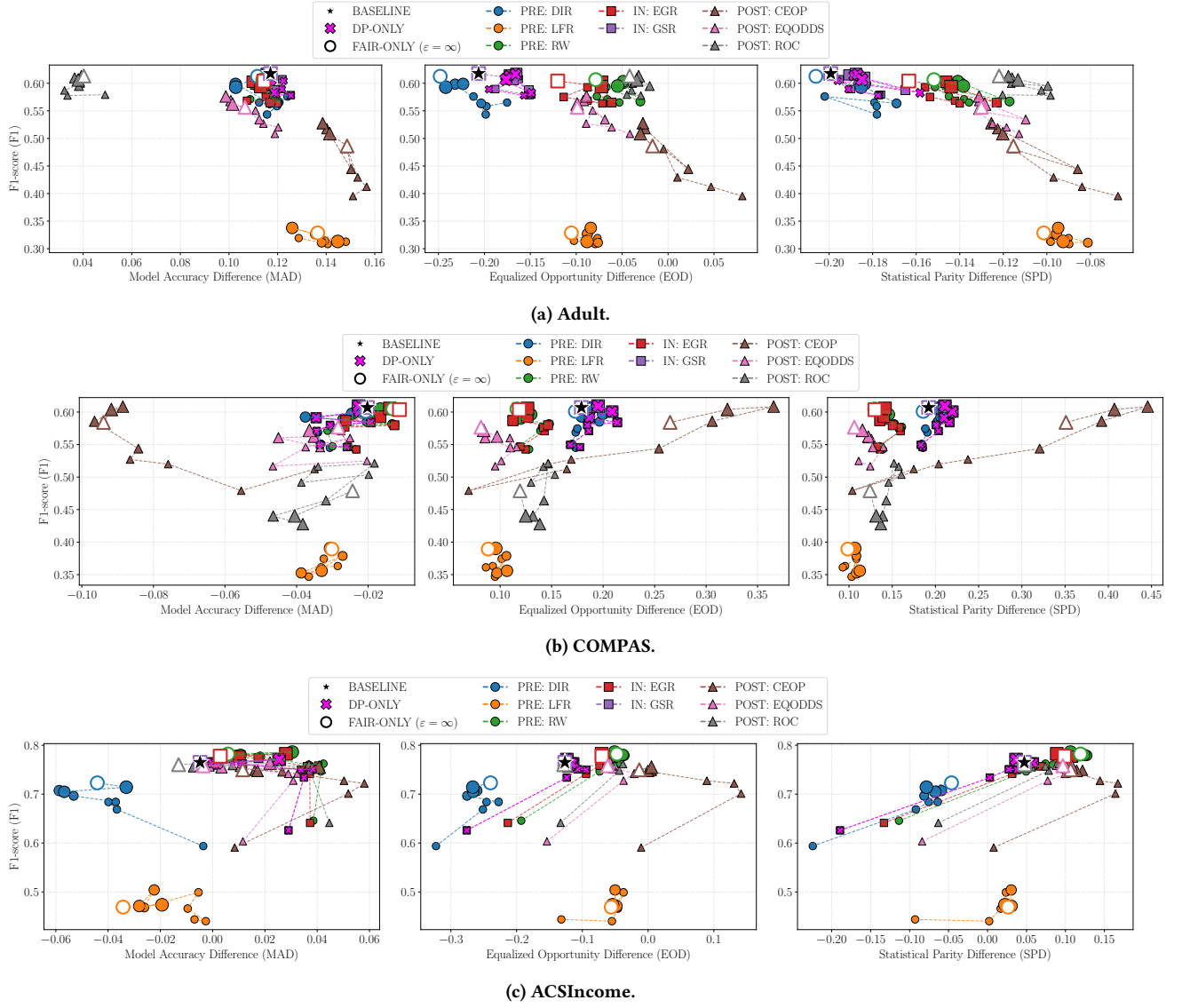


Figure 17: F1-score–fairness trade-offs under the AIM synthesizer across three real-world datasets. Each subfigure reports F1-score versus three fairness metrics (MAD, EOD, SPD) for each fairness intervention applied at the pre-, in-, and post-processing (with DP calibration set) stages. Marker size encodes the privacy budget ϵ for both DP-only and DP+Fair settings, while hollow markers indicate the Fair-only setting ($\epsilon = \infty$). The Baseline setting (no DP and no fairness intervention) is indicated by the $\star$ symbol.

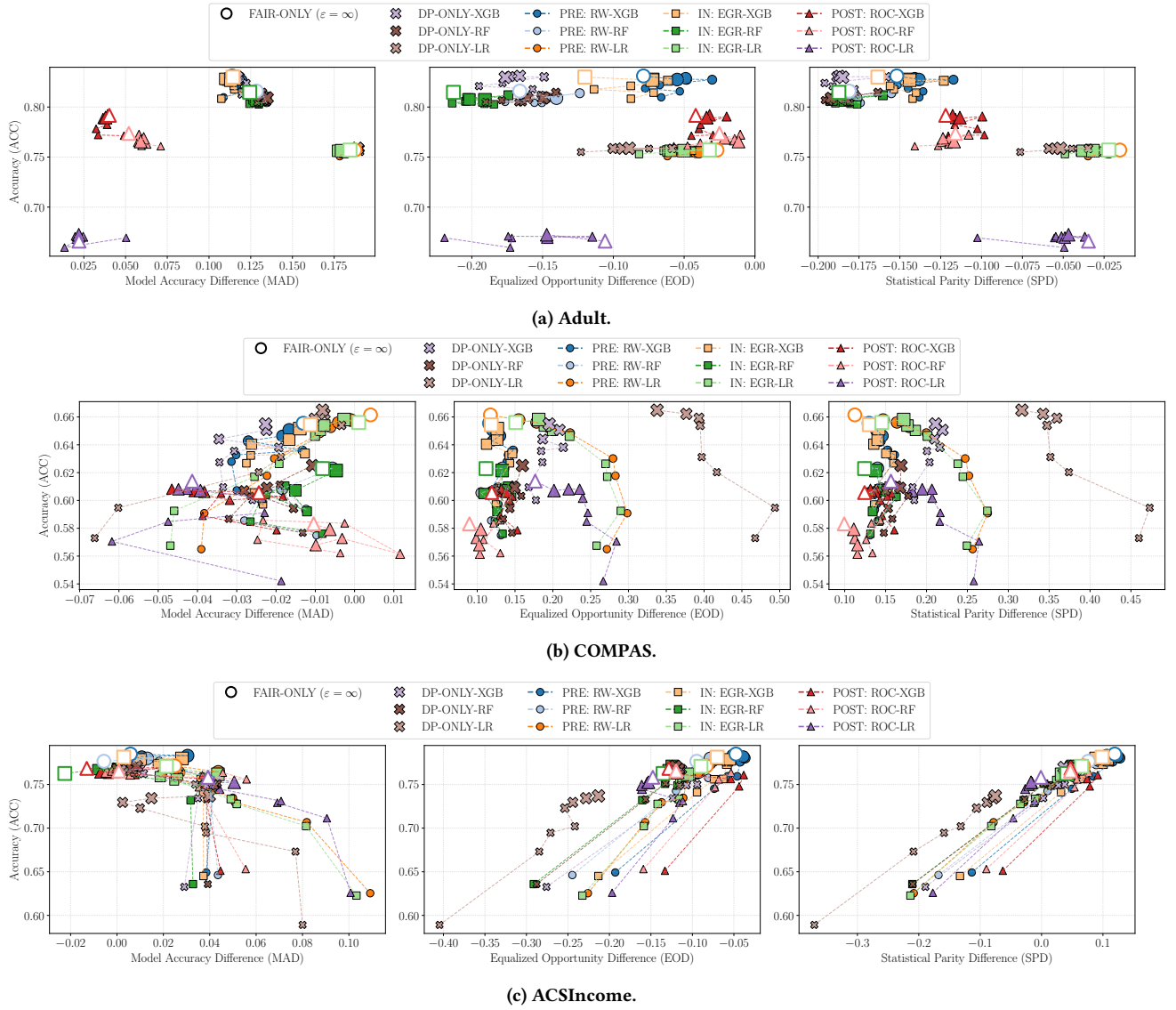


Figure 18: Accuracy–fairness trade-offs under the AIM synthesizer across three real-world datasets and three different machine learning models. Each subfigure reports accuracy (ACC) versus three fairness metrics (MAD, EOD, SPD) for each fairness intervention applied at the pre-, in-, and post-processing (with DP calibration set) stages. Marker size encodes the privacy budget ϵ for both DP-only and DP+Fair settings, while hollow markers indicate the Fair-only setting ($\epsilon = \infty$).